Dynamics of motor-load combination

(Mechanical Transmission System (MTS)

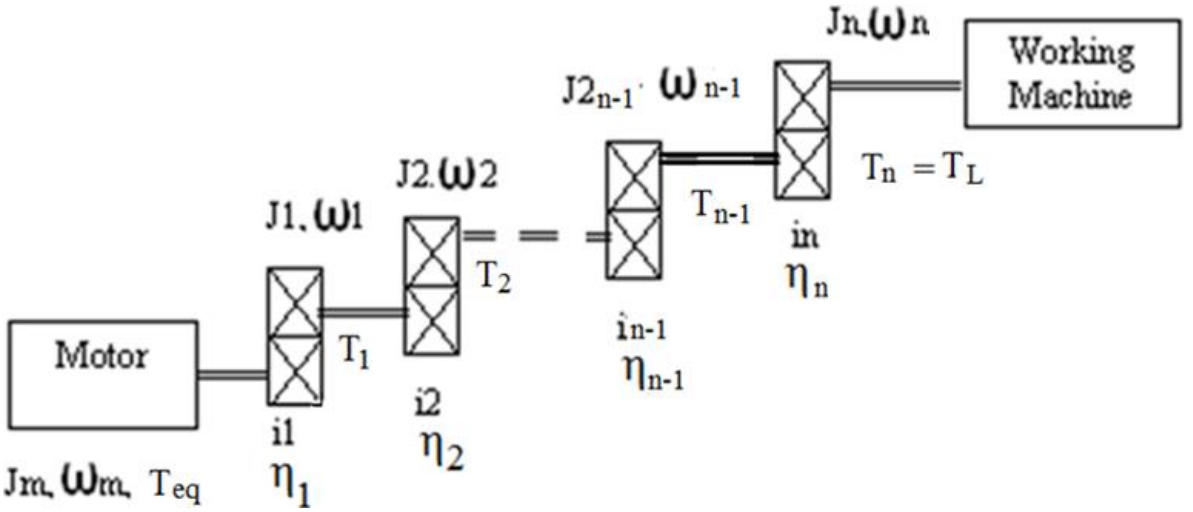
Different parts of a load may be coupled through different mechanisms, such as gears, V-belts, chains, sheaves and crankshafts. These parts may have different speeds and different types of motions such as rotational and translational.

Loading with rotational motion

$J_m, J_1, J_2 .. J_n$ – moment of inertia of motor, first transmission element, second ……. and load (nth-element) measured in Kg.m^2.

$\omega_m, \omega_1, \omega_2 \ldots \omega_n$ – speed of the motor, first transmission element, second ……. and load.

$$i_1 = \frac{\omega_m}{\omega_1}, i_2 = \frac{\omega_1}{\omega_2}, \ldots . i_n = \frac{\omega_{n-1}}{\omega_n}$$ - transmission ratios.



Equivalent values of MTS parameters

- **Transmission of the load torque.** If we assume that power at the loads and the motor is the same, the transmission efficiencies of gears are η_1 , $\eta_{2, \ldots.}$ η_n .

Equivalent or referred torque at the motor shaft, can be calculated as :

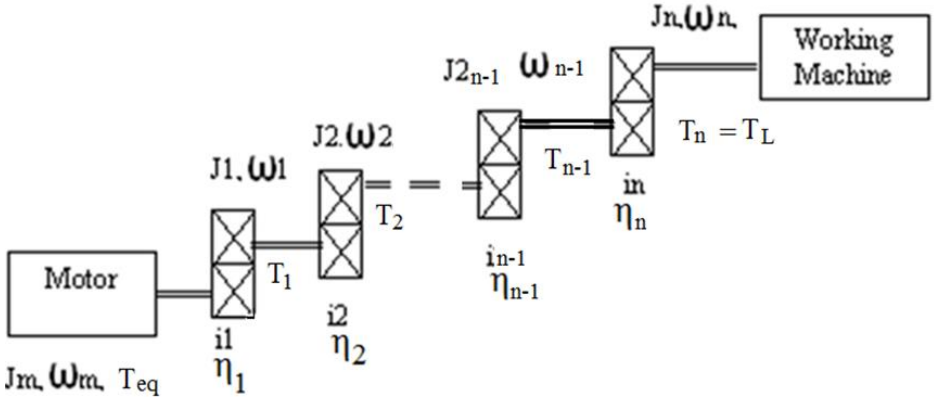
$$T_{ref} = \frac{1}{\eta_1 i_1} \times \frac{1}{\eta_2 i_2} \times \ldots \times \frac{T_n}{\eta_n i_n} = \frac{T_L}{\eta i}$$

Derivation of referred torque

$$\eta_n T_{n-1}\omega_{n-1} = T_n\omega_n \Rightarrow T_{n-1} = \frac{T_n}{\eta_n i_n}$$

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$$\eta_2 T_1\omega_1 = T_2\omega_2 \Rightarrow T_1 = \frac{T_2}{\eta_2 i_2}$$



$$\eta_1 T_{ref}\omega_m = T_1\omega_1 \Rightarrow T_{ref} = \frac{T_1}{\eta_1 i_1} = \frac{1}{\eta_1 i_1} \times \frac{T_2}{\eta_2 i_2}$$

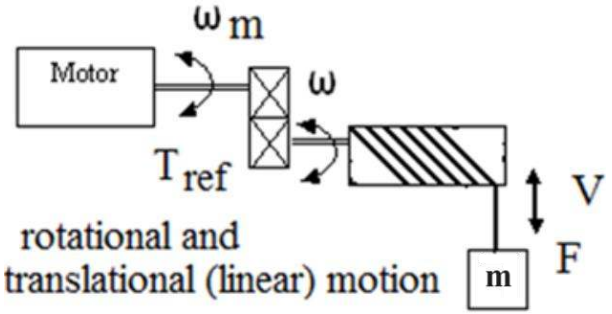
$$= \frac{1}{\eta_1 i_1} \times \frac{1}{\eta_2 i_2} \times \ldots \times \frac{T_n}{\eta_n i_n} = \frac{T_L}{\eta i}$$

Loading with translational motion

Transmission of load torque:

a. In the case of linear motion upward (lifting the weight)

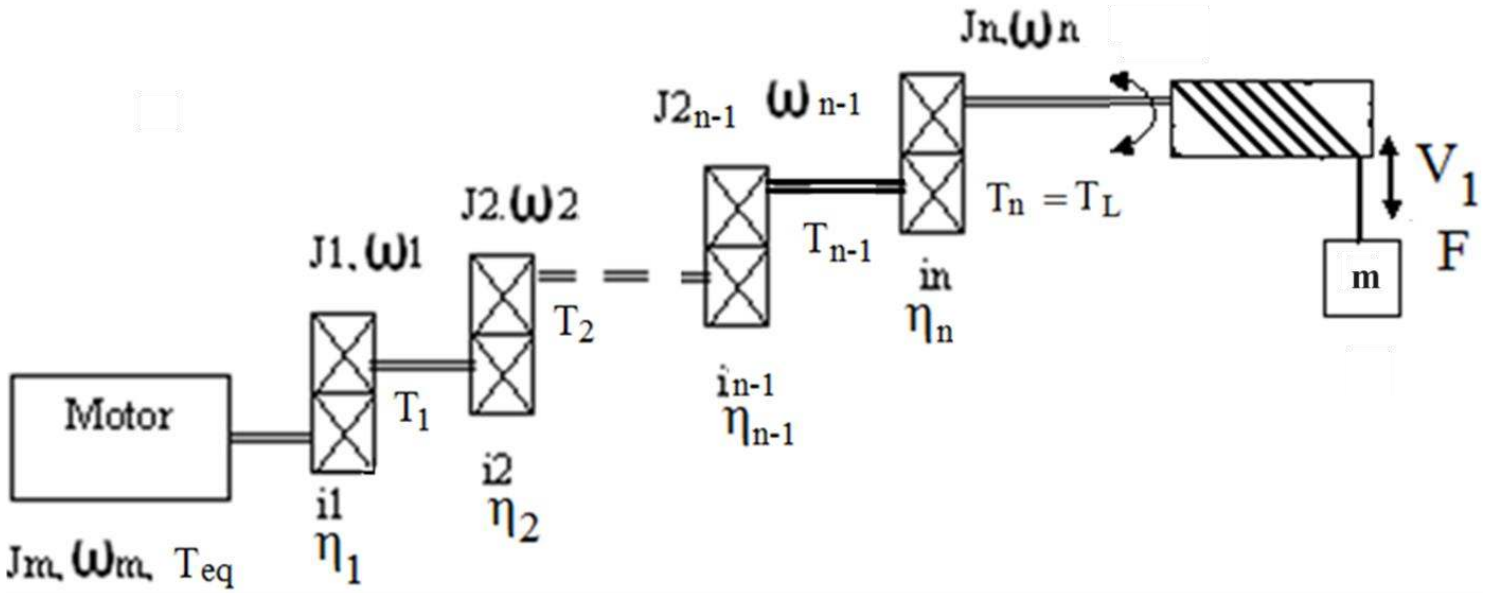
The power at the motor and the load should be the same



$$Fv = T_{ref}\omega_m\eta$$

For the translational system with many gears as in the figure below

$$T_{ref} = \frac{F}{\eta}\left(\frac{v}{\omega_m}\right) = \frac{F}{\eta}\frac{\omega R}{\omega_m} = \frac{FR}{(\omega_m/\omega)\eta} = \frac{T_L}{(i_1 i_2 \ldots i_n)(\eta_1\eta_2\ldots\eta_n)} = \frac{FR}{i\eta} = \frac{mgR}{i\eta}$$



b. In the case of linear motion downward (lowering the weight):

$$Fv\eta = T_{ref}\omega_m$$

$$T_{ref} = F\eta(\frac{v}{\omega_m}) = F\eta\frac{\omega R}{\omega_m} = \frac{\eta FR}{(\omega_m/\omega)} = \frac{(\eta_1\eta_2...\eta_n)T_L}{(i_1 i_2...i_n)} = \frac{\eta FR}{i} = \frac{\eta mgR}{i}$$

- where, R - Drum radius in m, m – load mass in kg

V=ωR- rope speed in m/s.

The functions of the mechanical transmission system (MTS) in drive systems are : 1- to step down the speed to meet the load speed. 2- to increase the torque to meet the required load torque. 3- to change the direction of motion. 4- to make suitable coupling between the motor and the load.

Equivalent values of MTS parameters

- **Transmission of the moment of inertia :**

If the losses in transmission are neglected, then the kinetic energy due to the equivalent inertia must be the same as the kinetic energy of various moving parts. Thus

a. For load that has rotational motion:

$$J_{ref}\omega_m^2/2 = J_m\omega_m^2/2 + J_1\omega_1^2/2 + ... + J_n\omega_n^2/2$$

So,

$$J_{ref} = J_m + J_1(\frac{\omega_1}{\omega_m})^2 + J_2(\frac{\omega_2}{\omega_m})^2 + ...J_n(\frac{\omega_n}{\omega_m})^2 = J_m + J_1\frac{1}{i_1^2} + J_2\frac{1}{i_1^2 i_2^2} + ... + J_n\frac{1}{i_1^2 i_2^2 ... i_n^2}$$

b. For load that has translational motion:

So, $J\ \omega_m^2/2 = m v^2/2$

$$J = m\left(\frac{V}{\omega_m}\right)^2 = m\left(\frac{R\omega_L}{\omega_m}\right)^2 = m\left(\frac{R}{\omega_m/\omega_L}\right)^2 = m\frac{R^2}{i^2}$$

c. For the MTS of two motion load elements (rotational and translational):

$$J_{ref}\frac{\omega_m^2}{2} = J_m\frac{\omega_m^2}{2} + J_1\frac{\omega_1^2}{2} + ... + J_n\frac{\omega_n^2}{2} + m\frac{v_1^2}{2}$$

$$J_{ref} = J_m + J_1/i_1^2 + J_2/i_1^2 i_2^2 + ... + J_n/i_1^2 i_2^2 .... i_n^2 + m(v/\omega_m)^2 =$$

$$J_{ref} = J_m + J_1/i_1^2 + J_2/i_1^2 i_2^2 + ... + J_n/i_1^2 i_2^2 .... i_n^2 + m(R\omega_n/\omega_m)^2 =$$

$$J_{ref} = J_m + J_1/i_1^2 + J_2/i_1^2 i_2^2 + ... + J_n/i_1^2 i_2^2 .... i_n^2 + m(R/i_1 i_2 ... i_n)^2 =$$

$$J_{ref} = J_m + J_1/i_1^2 + J_2/i_1^2 i_2^2 + ... + J_n/i^2 + mR^2/i^2$$

Problem 1:

Calculate the equivalent load torque and total moment of inertia referred to motor shaft in fig(1).

MTS parameters:

$\eta_m = 0.96$; $J_m = 1$ kg.m^2; $\omega_m = 100$ rad/sec.

$\eta_1 = 0.95$; $J_1 = 2$ kg.m^2 ; $\omega_1 = 5$ rad/sec.

$\eta_2 = 0.96$; $J2 = 1.5$ kg.m^2; $R = 0.3$ m ; $\omega_2 = 5$ rad/sec.

$m = 500$ kg; $g_m = 9.80665$ m/sec^2

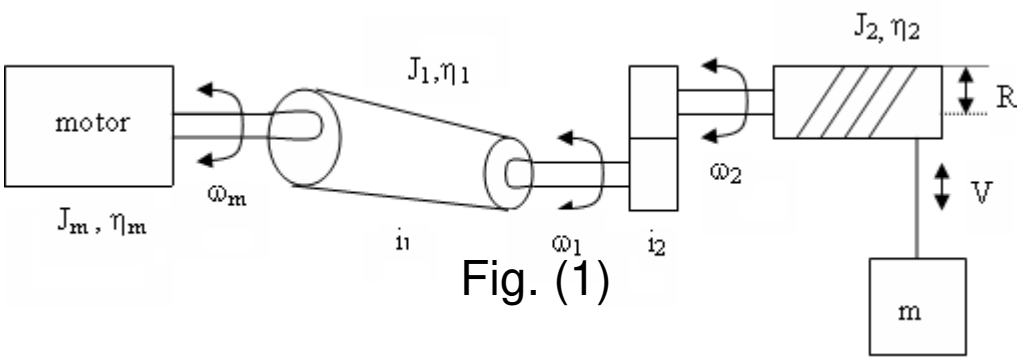


Fig. (1)

solution:

- 1- linear motion upward (lifting the weight)

$$J_\Sigma = J_M + J_1\left(\frac{1}{i_1^{\,2}}\right) + J_2\left(\frac{1}{i^2}\right) + m\frac{R^2}{i^2}, so \qquad J_\Sigma = 1 + 2\left(\frac{1}{20^2}\right) + 1.5\left(\frac{1}{20^2}\right) + 500\frac{0.3^2}{20^2} = 1.12125 kg.m^2$$

$$\eta_\Sigma = \eta_M.\eta_1.\eta_2 = (0.96)(0.95)(0.96) = 0.8664 = 86.64\%$$

$$T_{eq} = \frac{F.R}{i}.\frac{1}{\eta_\Sigma} = \frac{m.g.R}{i.\eta_\Sigma} = \frac{(500)(9.80665)(0.3)}{(20)(0.8664)} = 84.89 N.m$$

- 2- linear motion downward (lowering the weight)

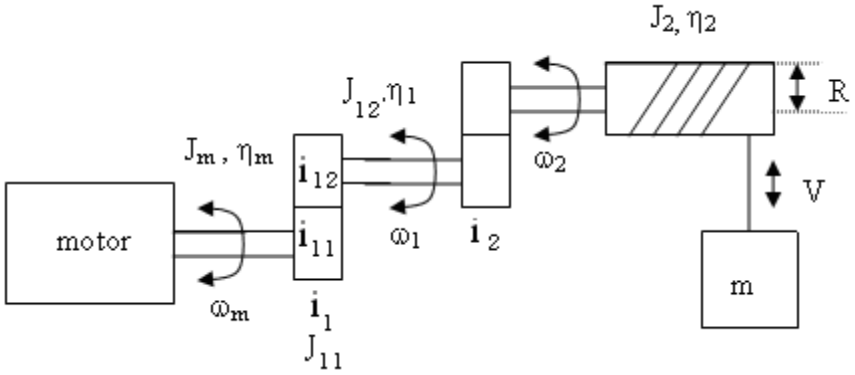
$$T_{eq} = \frac{F.R}{i}.\eta_\Sigma = \frac{m.g.R}{i}.\eta_\Sigma = \frac{(500)(9.80665)(0.3)(0.8664)}{(20)} = 63.72 N.m$$

- Solve problem 1, taking in consideration, that the gear unit consists of two parts (fig2)such that the moment of inertia of each gear is:

$$J_{11} = 0.5kg.m^2 \qquad J_{12} = 1.5kg.m^2$$

$$J_\Sigma = J_M + J_{11} + J_{21}\left(\frac{1}{i_1^2}\right) + J_2\left(\frac{1}{i^2}\right) + m\frac{R^2}{i^2} =$$

$$1 + 0.5 + 1.5\left(\frac{1}{20^2}\right) + 1.5\left(\frac{1}{20^2}\right) + 500\frac{0.3^2}{20^2} = 1.62kg.m^2$$



- in the case of linear motion and lifting the weight upward:

$$T_{eq} = \frac{F.R}{i}.\frac{1}{\eta_\Sigma} = \frac{m.g.R}{i.\eta_\Sigma} = \qquad \frac{(500)(9.80665)(0.3)}{(20)(0.8664)} = 84.89N.m$$

- But in the case of linear motion and lowering the weight downward:

$$T_{eq} = \frac{F.R}{i}.\eta_\Sigma = \frac{m.g.R}{i}.\eta_\Sigma = \qquad \frac{(500)(9.80665)(0.3)(0.8664)}{(20)} = 63.72N.m$$